

Quick Start Guide

Get up and running with RAG in 5 minutes!

Prerequisites

- Python 3.10 or higher
- pip package manager
- (Optional) OpenAI API key for generation

Installation

1. Clone or Navigate to Repository

```
cd /path/to/rag_tutorial
```

2. Create Virtual Environment

```
# Create virtual environment (tested with Python 3.13)
python -m venv .venv

# Activate it
source .venv/bin/activate # On Windows: .venv\Scripts\activate
```

3. Install Dependencies

```
# Install core dependencies
pip install -r requirements.txt

# This will install:
# - chromadb (vector database)
# - sentence-transformers (embeddings)
# - openai (for GPT models)
# - other utilities
```

4. Set Up Environment Variables (Optional)

```
# Copy example file
cp .env.example .env

# Edit .env and add your OpenAI API key
# (Not required for basic examples)
```

Running Examples

Example 1: Simple QA (No API Key Required)

```
cd examples/01-simple-qa
python simple_qa.py
```

This example works without an API key and demonstrates basic RAG concepts.

Example 2: Document Chat (Requires API Key)

```
# Set your API key first
export OPENAI_API_KEY="your-key-here"

cd examples/02-document-chat
python document_chat.py
```

This provides an interactive chat interface with your documents.

Testing Your Setup

Quick test to verify everything works:

```
python -c "  
from sentence_transformers import SentenceTransformer  
import chromadb  
  
print('Loading embedding model...')  
model = SentenceTransformer('all-MiniLM-L6-v2')  
print('✓ Embeddings working')  
  
print('Testing ChromaDB...')  
client = chromadb.Client()  
collection = client.create_collection('test')  
collection.add(documents=['test'], ids=['1'])  
print('✓ Vector database working')  
  
print('\nSetup successful! You are ready to learn RAG.')  
"
```

Learning Path

1. **Read the lessons** in order, starting with [lessons/01-introduction-to-rag.md](#)
2. **Run the examples** to see concepts in action
3. **Experiment** with the utility functions in [utils/](#)
4. **Build your own** RAG system!